

# Food-safety regulation case: extracting a nutrient-content claim

FoodReg test set · complex indicator value (IV) with a comparison target and percentage constraint

## FoodReg test clause (translated)

Nutrient condition: increase in minerals (excluding sodium).

**Gold indicator value (IV):** Compared with reference food, mineral content increases by  $\geq 25\%$  (inclusive).

The highlighted span combines a reference object, a nutrient component, and a regulatory threshold.

## Full SPSR-Net

Exact gold IV span retained

**$p(\text{IV}) = 0.947$**

$>$  decision threshold  $\tau = 0.48$

Correct extraction of the nutrition claim

## w/o SNSA

Exact gold IV span rejected

**$p(\text{IV}) = 0.146$**

$<$  decision threshold  $\tau = 0.48$

False negative for the nutrition claim

Interpretation: on this real food-safety rule, SNSA changes the exact-span decision from a missed IV to a retained IV.

Scores are sigmoid probabilities for the exact gold span from the saved Full and w/o SNSA checkpoints; this is a case study, not an aggregate metric.